Siddharth Chaini

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Education

University of Delaware — **Newark, DE, USA**

Ph.D. in Physics and Astronomy *(GPA: 3.96/4.0)* — Sep. 2022 – Apr. 2027 (expected)

NASA FINESST Fellow & **UD Doctoral Fellow for Excellence**

Advisor: Federica B. Bianco

Dissertation: *"Discovering Structure and Surprise in Time-Domain Astrophysics"*

Data-driven discovery pipelines for next-generation surveys focusing on out-of-distribution anomaly detection of transients & variables, data-driven modelling of transients via reconstruction of sparse multisurvey & multifilter light curves, and physical characterization of supernovae spectra via data-driven methods.

Indian Institute of Science Education and Research (IISER) Bhopal — **Bhopal, India**

BS-MS (Dual Degree) in Physics *(CPI: 9.04/10)* — Aug. 2017 – May 2022

Publications

* denotes equal contribution; >3 authors shortened to "et al. (incl. **S. Chaini**)".

Lead & Major-Contribution Publications

[13] **S. Chaini**, W. F. Fortino, S. Daniels, et al., "Data-Driven Representations of Supernovae Spectra: From Physical Features to Evolution." ***In Prep.***

[12] N. Earl, **S. Chaini**, K. D. French, et al., "NAPTIME: A Neural-Process Framework for Rubin Alert Classification," ApJ, vol. 1010, no. 1, p. 11, Oct. 2026. DOI: 10.3847/1538-4357/aea0d1. arXiv: 2607.19236.

[11] **S. Chaini**, F. B. Bianco, and A. Mahabal, "Probabilistic Data-Driven Modelling of Astrophysical Transients: The Neural Process Family for Ultrafast and Class-Agnostic Light Curve Reconstruction with NightLANP," May 2026. arXiv: 2605.27527. ***Accepted, NeurIPS 2026 - main track.***

Also accepted as a workshop paper at ICML 2026, AI for Physics [OpenReview].

[10] **S. Chaini**, F. B. Bianco, and A. Mahabal, "In Search of the Unknown Unknowns: A Multi-Metric Distance Ensemble for Out of Distribution Anomaly Detection in Astronomical Surveys," in *NeurIPS 2025 Workshop on Machine Learning and the Physical Sciences (ML4PS)*, Oct. 2025. arXiv: 2510.23702.

[9] B. Domínguez*, **S. Chaini***, K. Mrakovčić*, et al., "Validation of the Bond et al. (2010) SDSS-derived kinematic models for the Milky Way's disk and halo stars with Gaia Data Release 3 proper motion and radial velocity data," A&A, vol. 688, A132, Aug. 2024, Art. no. A132. DOI: 10.1051/0004-6361/202449653. arXiv: 2406.03541.

[8] **S. Chaini**, A. Mahabal, A. Kembhavi, et al., "Light curve classification with DistClassiPy: A new distance-based classifier," *Astronomy and Computing*, vol. 48, p. 100 850, Jul. 2024, Art. no. 100850. DOI: 10.1016/j.ascom.2024.100850. arXiv: 2403.12120.

[7] **S. Chaini**, A. Bagul, A. Deshpande, et al., "Photometric identification of compact galaxies, stars, and quasars using multiple neural networks," MNRAS, vol. 518, no. 2, pp. 3123–3136, Jan. 2023. DOI: 10.1093/mnras/stac3336. arXiv: 2211.08388.

[6] **S. Chaini** and S. S. Kumar, "Astronomical Classification of Light Curves with an Ensemble of Gated Recurrent Units," Jun. 2020. arXiv: 2006.12333.

Other Co-Authored Publications

[5] D. J. Sand, B. Subrayan, C. L. Ransome, et al. (incl. **S. Chaini**), "Nearby Supernova Science in the Era of High Cadence, Deep Time Domain Surveys," Sep. 2026. arXiv: 2609.38304. ***To be submitted.***
The Shadow Collaboration.

[4] C. R. Bom, A. Mahabal, F. Bianco, et al. (incl. **S. Chaini**), "Toward decision-aware AI for LSST-scale time-domain astronomy," Jun. 2026. arXiv: 2606.05285. ***Under review, Nature Astronomy.***
Rubin LSST Transients and Variable Stars Science Collaboration.

[3] E. J. Honaker, J. E. Gizis, C. Aganze, et al. (incl. **S. Chaini**), "Searching for Ultracool Dwarfs in Early LSST Data Products," ApJ, vol. 1004, no. 2, p. 214, Jun. 2026, Art. no. 214. DOI: 10.3847/1538-4357/ae6ce9. arXiv: 2604.08387.

[2] C. L. Ransome, B. M. Subrayan, D. J. Sand, et al. (incl. **S. Chaini**), "First Results from the LSST Shadow Survey: The Restless Luminous Blue Variable AT2017des in the Virgo-Cluster Galaxy, NGC4532," ApJL, Jun. 2026. arXiv: 2606.23784. ***Under review, ApJL.***
The Shadow Collaboration.

[1] A. Crawford, F. Bianco, T. A. Pritchard, et al. (incl. **S. Chaini**), "Quoth the Maven: Testing a Foundation AI Model on the Classification of Single- versus Double-peaked Stripped-envelope Supernova Light Curves," 2026. ***To be.***

Software

NightLANP — github.com/sidchaini/NightLANP (2026).
keras-neural-processes — github.com/sidchaini/keras-neural-processes (2024).
DistClassiPy — github.com/sidchaini/DistClassiPy (2024); ascl:2403.002.

Technical Skills

Surveys & data	Rubin/LSST (DP0–2; Alerts: ALeRCE, Fink), DECam, ZTF, TESS, SDSS, Gaia
Languages	Python, C, C++, Java, Wolfram Language, SQL, HTML, Bash, Assembly, LaTeX
Libraries	PyTorch, TensorFlow/Keras, scikit-learn, Astropy, NumPy, pandas, Matplotlib, Selenium
Tools	Git, Jupyter, VS Code, Mathematica, SAOImage DS9, Aperture Photometry Tool

Fellowships, Grants and Awards

- **NASA FINESST** — Grant 80NSSC25K0312; **US $100,000** 2025 – 2027
  Future Investigator: Enabling Serendipitous Discovery with Machine Learning for Large Sky Surveys: Synergizing TESS, Roman, and Rubin; 24/456 Astrophysics proposals selected (∼5%)
- **NSF Research Grant** — Grant 2511639; **US $395,000** 2025 – 2028
  Co-author on the funded proposal (PI: F. B. Bianco); graduate student researcher
- **Doctoral Fellowship for Excellence** — University of Delaware; **US $39,000** 2025 – 2026
  2 awarded per year in Dept. of Physics & Astronomy
- LINCC Frameworks Community Event Support, *LSST Discovery Alliance* 2026
- Data Science Institute Fellow, *University of Delaware* 2023 – 2027
- Vatican Observatory Scholarship, *18th Summer School* – 25 selected from ∼200 applicants, 20 countries 2023
- INSPIRE Scholarship, *DST, Government of India* – Top 1% of ∼1.2 million applicants nationally 2017 – 2022

Observing Experience

- **Dark Energy Camera (DECam)**, Cerro Tololo Inter-American Observatory – Remote observer, DECam Shadow project 2 nights

Academic Leadership & Service

- **Elected Member**, LSST ISSC Executive Council Jun. 2026 – Present
  Informatics and Statistics Science Collaboration — oversight of working groups, committee appointments and collaboration policies
- **Co-Chair**, LSST TVS Anomalies Subgroup Sep. 2024 – Present
  Rubin LSST Transients and Variable Stars Science Collaboration — organize meetings and coordinate subgroup activities
- **Organizing Committee**, *LSST is here!* DP2 Hackathon Aug. 2026
  Co-organized a 2-day hackathon on Rubin Observatory's LSST Data Preview 2 (DP2) data and tools
- **Instructor**, Rubin TVS Software Taskforce – Multiple collaboration-wide software workshops Aug. 2026
- **Reviewer**, workshop programme committees – ICML Workshop on AI for Physics; NeurIPS Workshop on Representations for Physical Sciences 2026
- **Elected Head**, Student Research Group, IISER Bhopal Astronomy Club Aug. 2020 – Aug. 2021
  In charge of group activities and data analysis — image reduction and CCD photometry

Talks, Panels & Posters

Invited:

- Quasar Bazaar Hackweek, SISSA, Trieste, Italy (Talk) Feb. 2026
  Distance Multi-Metric Anomaly Detection for LSST Transients & Variables
- Astronomy Seminar, National Centre for Radio Astrophysics, Pune, India (Talk) Jan. 2026
  Transient hunting in the era of large-sky surveys with Neural Processes
- AI/ML Applications in Astronomy and Astrophysics, IUCAA, Pune, India (Panelist) Jan. 2025
  Pitfalls of using ChatGPT
- LSST TVS Colloquium, Online (Talk) Oct. 2024
  Distance Metrics for Machine Learning in Time-Domain Astronomy with DistClassiPy

Contributed:

- Physics in Supernovae Spectral Manifolds
  European Astronomical Society 2026 (poster).
- Probabilistic Light Curve Modelling with NightLANP
  Rubin Community Workshop 2026 (talk), Extragalactic Transient Universe (talk), European Astronomical Society 2026 (talk), APS Global Physics Summit 2026 (talk), TDABench 2026 (talk), LSST@Europe 7 (talk), IAU Symposium 406 (poster).
- Distance Multi-Metric Anomaly Detection
  Rubin Community Workshop 2025 (talk), NeurIPS: ML and the Physical Sciences 2025 (poster), Shaping the Future of Time-Domain Astronomy with LSST (talk).
- Light Curve Classification with DistClassiPy
  AI/ML Applications in Astronomy and Astrophysics (tutorial), LSST@Europe 6 (talk), Rubin Community Workshop 2024 (talk), Rare Gems in Big Data 2024 (poster), APS Mid-Atlantic Section 2023 (poster).

Mentoring

- Co-mentored three Lincoln University undergraduates — May 2026 – Aug. 2026
  Abdourahim Sylla, Agustina Lorda and Adeoluwa Afolabi
- Co-mentored UD undergraduate NASA Space Grant Fellow — Jul. 2023 – Aug. 2024
  Ibrahim Wilson

Teaching Experience

- **Lesson Instructor**, Rubin LSST TVS Science Collaboration — Jul. 2024 – Present
  Intermediate Python for Astronomical Software Development; Research Software Engineering Mini Course
- **Teaching Assistant**, University of Delaware — Sep. 2022 – May. 2025
  PHYS 207/227: Fundamentals of Physics; PHYS 208: Electromagnetism; SCEN 101: Physical Science
- **Teaching Assistant**, IISER Bhopal — Jan. 2019 – May 2019
  ECS 102: Introduction to Programming

Outreach

- **DP2 Hackathon**, University of Delaware — co-organized a hackathon targeted primarily at ECRs in the Mid-Atlantic region to share and work on Rubin DP2 science together.
- **Rubin First Look**, University of Delaware — co-led a public viewing party for Rubin Observatory's first images, and presented LSST's goals and science to the public.
- **Astronomy Club**, IISER Bhopal — organized telescope viewing nights and computational-astronomy programming tutorials for the campus community.
- **Outreach talks**, tutoring centres in Thane, India — science talks to students in grades 8–10 to spark interest in astrophysics.
- **Media coverage** — *UDaily*: "NASA's next generation of scientists" (Oct. 2025), "Opening a new window into the Universe" (Jun. 2025); *Franciscan Media*: "Vatican hosts astronomers of tomorrow" (Jun. 2023, also on YouTube).

References

- **Federica B. Bianco** (University of Delaware) — *PhD advisor* — fbianco@udel.edu
- **Ashish Mahabal** (Caltech) — *MS thesis co-advisor* — aam@astro.caltech.edu
- **Željko Ivezić** (Rubin Observatory/UW) — *mentor* — ivezic@uw.edu